SA1 - Week 1 Report

jp2056

May 2026

1 Point vortex stream function

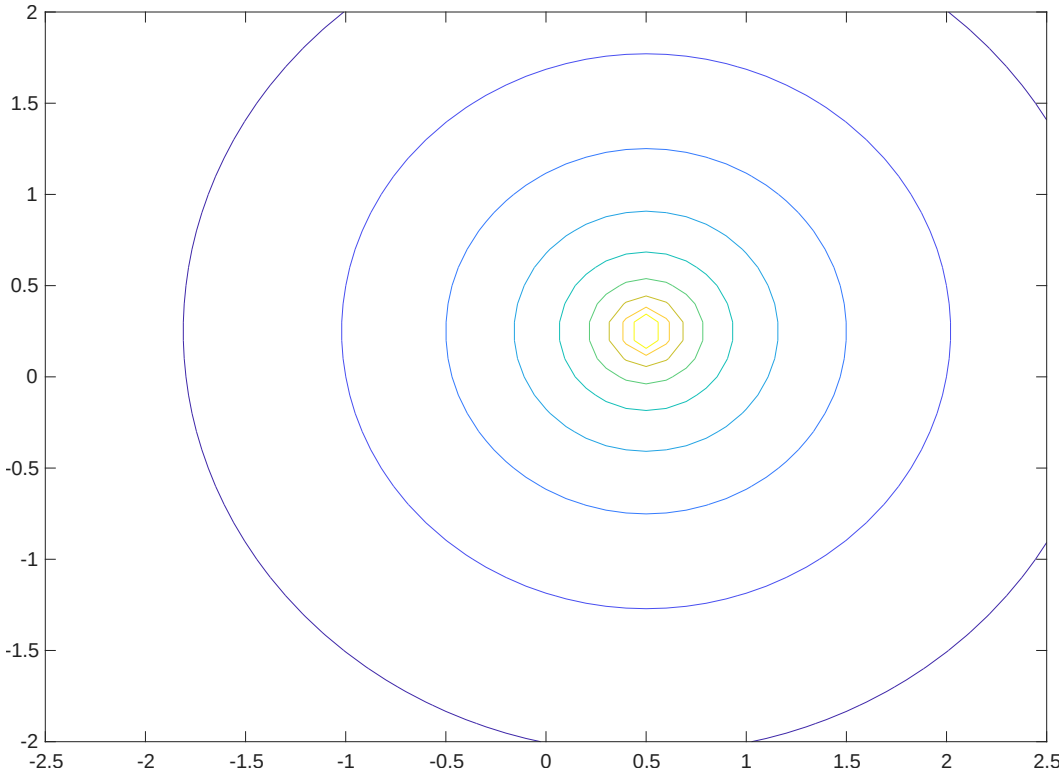


Figure 1: Stream function plot for a point vortex. (Exercise 1)

Listing 1: The function to evaluate the stream function due to a point vortex at an arbitrary point. (Exercise 1)

```
function psixy = psipv(xc, yc, Gamma, x, y)
    %x and y can be matricies
    %find the elementwise squared distance
    r_2 = (x-xc).^2 + (y-yc).^2;
    %evaluate the streamfunction elementwise
    psixy = - (Gamma / (4 * pi)) * log(r_2);
end
```

Listing 2: The script for plotting the stream function of a point vortex with $\Gamma = 3$ and $(x_c, y_c) = (0.5, 0.25)$. (Exercise 1)

```
xmin = -2.5; xmax = 2.5;
ymin = -2;   ymax = 2;
nx = 51; ny = 41;
Gamma = 3.0;
xc = 0.5; yc = 0.25;

x = linspace(xmin, xmax, nx);
```

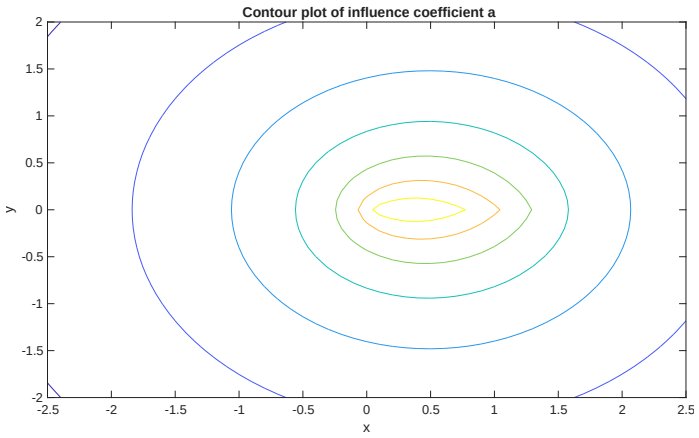
```
y = linspace(ymin, ymax, ny);

%create a matrix of x, and matrix of y coordinates
[xm, ym] = meshgrid(x, y);

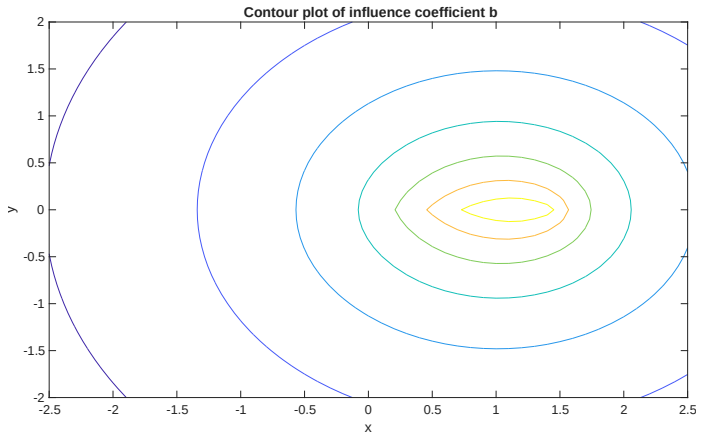
%evaluate the streamfunction at each point
psi = psipv(xc, yc, Gamma, xm, ym);

c = -0.4:0.2:1.2;
contour(xm, ym, psi, c);
```

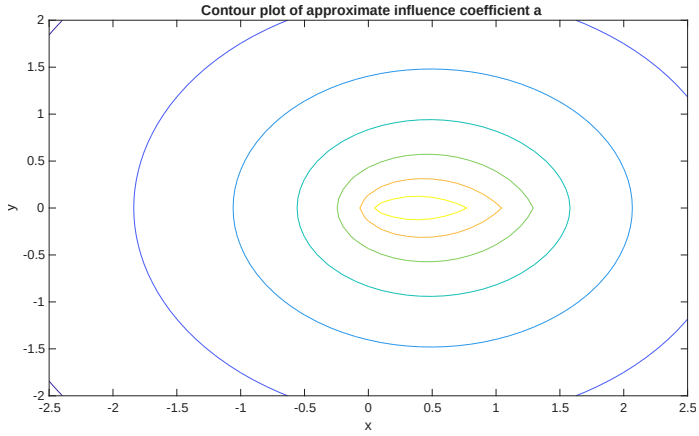
2 Reference vortex-sheet-panel stream function



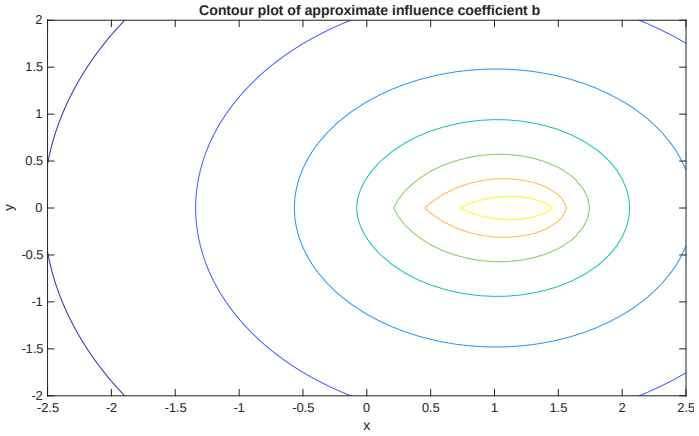
(a) Influence Factor A, f_a



(b) Influence Factor B, f_b



(c) Approximate Influence Factor A, f_a



(d) Approximate Influence Factor B, f_b

Figure 2: Reference vortex sheet panel stream function plots.

Listing 3: Function for evaluating influence coefficients f_a, f_b at an arbitrary point due to a reference vortex sheet which starts at $(0, 0)$, has length Δ and is along the x-axis. (Exercise 2)

```
function [infa, infb] = refpainf(del, X, Yin)
    if abs(Yin) <1e-6
        Y = 1e-6;
    else
        Y = Yin;
    end

    I0 = (-1/(4*pi)) * (X*log(X^2+Y^2) - (X-del)*log((X-del)^2 + Y^2) -2*del + ...
        2*Y*(atan(X/Y)-atan((X-del)/Y)));
    I1 = (1/(8*pi))*((X^2+Y^2)*log(X^2+Y^2) - ((X-del)^2+Y^2)*log((X-del)^2+Y^2)-2*X*del+del^2);
    infa = (1-(X/del))*I0 - I1/del;
    infb = (X/del)*I0 + I1/del;
end
```

Listing 4: Script for plotting the stream function of a reference vortex sheet panel . (Exercise 2)

```
clear;
close all;
xmin = -2.5;
xmax = 2.5;
ymin = -2;
ymax = 2;
ny = 41;
```

```
nx = 51;
del = 1.5;
gamma_a = 0.5;
gamma_b = 1.5;
nv=100;
yc=0;
psi_approx = zeros(nx, ny);
infa = zeros(nx,ny) ;
infb = zeros(nx, ny);
xm = zeros(nx, ny);
ym = zeros(nx, ny);
psi_analy = zeros(nx,ny);


for i = 1:nx
    for j = 1:ny
        xm(i,j) = xmin + (i-1)*(xmax-xmin)/(nx-1);
        ym(i,j) = ymin + (j-1)*(ymax-ymin)/(ny-1);
        [infa(i,j), infb(i,j)] = refpainf(del, xm(i,j), ym(i,j));
        psi_analy(i,j) = gamma_a*infa(i,j) + gamma_b *infb(i,j);

    end
end
for n = 1:nv
    xc = (2*n-1)*del/(2*nv);
    Gamma = (gamma_a*(del-xc)/del + gamma_b*xc/del)*del/nv;
    for i = 1:nx
        for j = 1:ny
            psi_approx(i,j) = psi_approx(i,j) + psipv(xc, yc, Gamma, xm(i,j), ym(i,j));
        end
    end
end

c = -0.15:0.05:0.15;

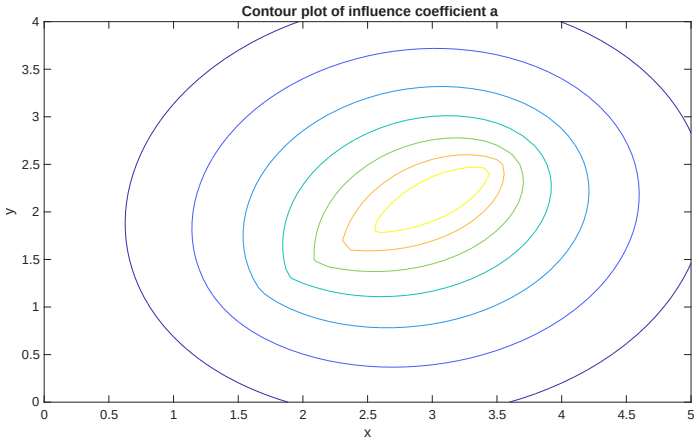
contour(xm,ym,infa,c);
xlabel('x');
ylabel('y');
title('Contour plot of influence coefficient a');

figure(2);
contour(xm,ym,infb,c);
xlabel('x');
ylabel('y');
title('Contour plot of influence coefficient b');

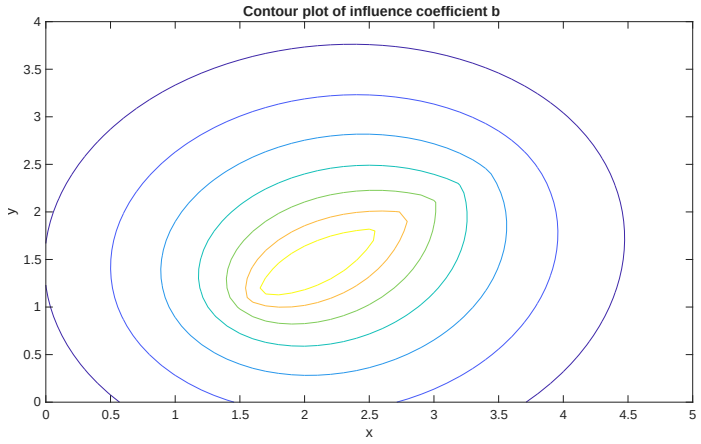
figure(3);
contour(xm,ym,psi_analy,c);
xlabel('x');
ylabel('y');
title('Contour plot of the analytical solution');

figure(4);
contour(xm,ym,psi_approx,c);
xlabel('x');
ylabel('y');
title('Contour plot of discretised approximation');
```

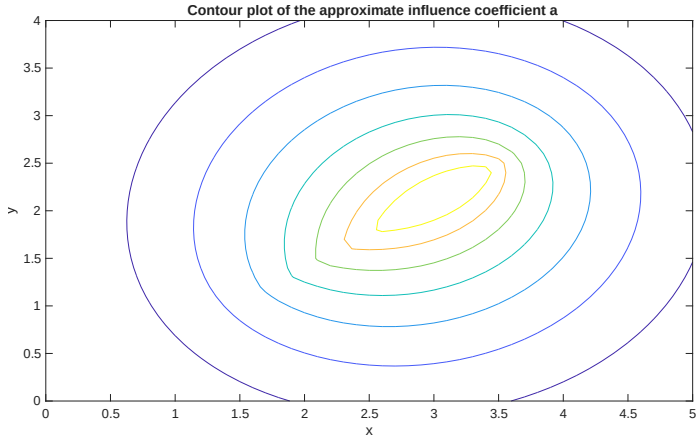
3 General vortex-sheet-panel stream function



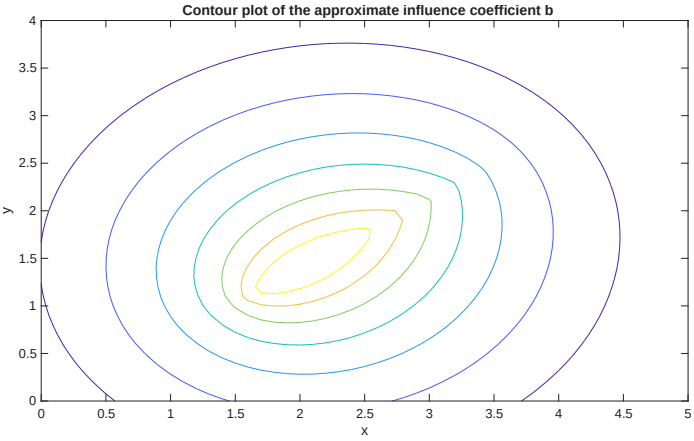
(a) Influence Factor A, f_a



(b) Influence Factor B, f_b



(c) Approximate Influence Factor A, f_a



(d) Approximate Influence Factor B, f_b

Figure 3: General vortex-sheet-panel stream function plots.

Listing 5: Function for evaluating the influence coefficients for any vortex-sheet-panel at an arbitrary point in space. (Exercise 3)

```
function [infa, infb] = panelinf(xa, ya, xb, yb, x, y)
    del = norm([(xb-xa), (yb-ya)]); %find the length of the sheet panel
    r = [(x-xa), (y-ya)]; %define the vector from the start of the sheet panel to the arbitrary point (x,y)
    t = [(xb-xa), (yb-ya)]/del; %find the unit vector tangential to the sheet panel
    n = [(yb-ya), -(xb-xa)]/del; %find the unit vector normal to the sheet panel
    X = dot(r, t); %find the tangential distance from the sheet panel
    Y = dot(r, n); %find the perpendicular distance from the sheet panel
    [infa, infb] = refpainf(del, X, Y); %evaluate the streamfunction using the reference sheet panel function
end
```

Listing 6: Script for plotting streamfunction for an inclined vortex-sheet-panel. (Exercise 3)

```
clear;
close all;
xmin = 0;
xmax = 5;
ymin = 0;
ymax = 4;
ny = 41;
nx = 51;
del = 1.5;
%gamma_a = 0.5;
```

```
%gamma_b = 1.5;
nv=100;
yc=0;
psi_approx = zeros(nx, ny);
infa = zeros(nx,ny) ;
infb = zeros(nx, ny);
xm = zeros(nx, ny);
ym = zeros(nx, ny);
psi_analy = zeros(nx,ny);
infa_approx = zeros(nx,ny);
infb_approx = zeros(nx,ny);
xa = 3.5;
ya = 2.5;
xb = 1.6;
yb = 1.1;

for i = 1:nx
    for j = 1:ny
        xm(i,j) = xmin + (i-1)*(xmax-xmin)/(nx-1);
        ym(i,j) = ymin + (j-1)*(ymax-ymin)/(ny-1);
        [infa(i,j), infb(i,j)] = panelinf(xa, ya, xb, yb, xm(i,j), ym(i,j));
        %psi_analy(i,j) = gamma_a*infa(i,j) + gamma_b *infb(i,j);

    end
end


l=0;
panel_length = norm([(xb-xa), (yb-ya)]);
dl = panel_length/nv;
pl_vec = [(xb-xa), (yb-ya)]/norm([(xb-xa), (yb-ya)]);
a = [xa,ya];
b = [xb, yb];
gamma_a = 1.0;
gamma_b = 0;

%evaluate approximate influence coefficient a using 100 point vorticies
for n = 1:nv
    %this code is designed to evaluate arbitrary values of gamma_a and
    %gamma_b but has been adapted to just evaluate influence a and
    %influence b
    centre = a + ((2*n-1)/2)*dl*pl_vec;
    Gamma = (gamma_a*norm(b-centre)/panel_length + gamma_b*norm(centre-a)/panel_length)*dl; %the interpolation
    of the two gammas to find an estimate of the point vortex size
    for i = 1:nx
        for j = 1:ny
            infa_approx(i,j) = infa_approx(i,j) + psipv(centre(1), centre(2), Gamma, xm(i,j), ym(i,j));
        end
    end
end

%evaluate approximate influence coefficient b using 100 point vcorticies
gamma_b = 1.0;
gamma_a = 0;
for n = 1:nv
    centre = a + ((2*n-1)/2)*dl*pl_vec;
    Gamma = (gamma_a*norm(b-centre)/panel_length + gamma_b*norm(centre-a)/panel_length)*dl;
    for i = 1:nx
        for j = 1:ny
            infb_approx(i,j) = infb_approx(i,j) + psipv(centre(1), centre(2), Gamma, xm(i,j), ym(i,j));
        end
    end
end

```

```
c = -0.15:0.05:0.15;

contour(xm,ym,infa,c);
xlabel('x');
ylabel('y');
title('Contour plot of influence coefficient a');
print -deps2c infa_ex3.eps;


figure(2);
contour(xm,ym,infb,c);
xlabel('x');
ylabel('y');
title('Contour plot of influence coefficient b');
print -deps2c infb_ex3.eps;


figure(3);
contour(xm,ym,infa_approx,c);
xlabel('x');
ylabel('y');
title('Contour plot of the approximate influence coefficient a');
print -deps2c infa_approx_ex3.eps;


figure(4);
contour(xm,ym,infb_approx,c);
xlabel('x');
ylabel('y');
title('Contour plot of the approximate influence coefficient b');
print -deps2c infb_approx_ex3.eps;
```

4 Cylinder flow streamlines

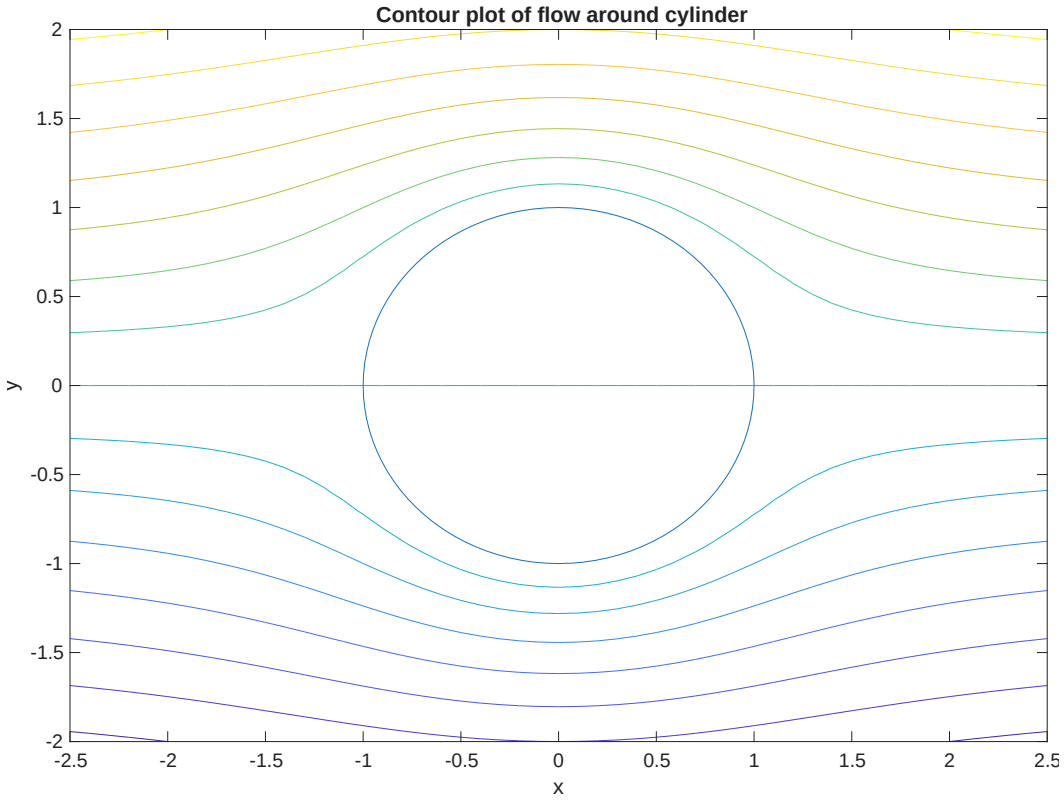


Figure 4: Streamlines around a cylinder (Exercise 4)

Listing 7: Script for producing the stream function plot of a cylinder. (Exercise 4)

```
clear;
xmin = -2.5;
xmax = 2.5;
ymin = -2;
ymax = 2;
ny = 41;
nx = 51;
del = 1.5;

psi = zeros(nx, ny);
xm = zeros(nx, ny);
ym = zeros(nx, ny);

np = 100;

%initalise the angles which will be used
theta = (0:np)*2*pi/np;

%establish the coordinates for the cylinder
for i = 1:np+1
    xs(i) = cos(theta(i));
    ys(i) = sin(theta(i));
end

%values of gamma_a and gamma_b around the cylinder for the different panels
gamma_panel_ends = -2 *sin(theta);

disp(gamma_panel_ends)

```

```
for i = 1:nx
    for j = 1:ny
        xm(i,j) = xmin + (i-1)*(xmax-xmin)/(nx-1);
        ym(i,j) = ymin + (j-1)*(ymax-ymin)/(ny-1);
        psi(i,j) = ym(i,j);
        for s = 1:np
            %evaluate the influence coefficients and contribution to streamfunction due to the vortex-sheet-
    panels
            [infa, infb] = panelinf(xs(s), ys(s), xs(s+1), ys(s+1), xm(i,j), ym(i,j));
            psi(i,j) = psi(i,j) + gamma_panel_ends(s)*infa + gamma_panel_ends(s+1) *infb;
        end
    end
end

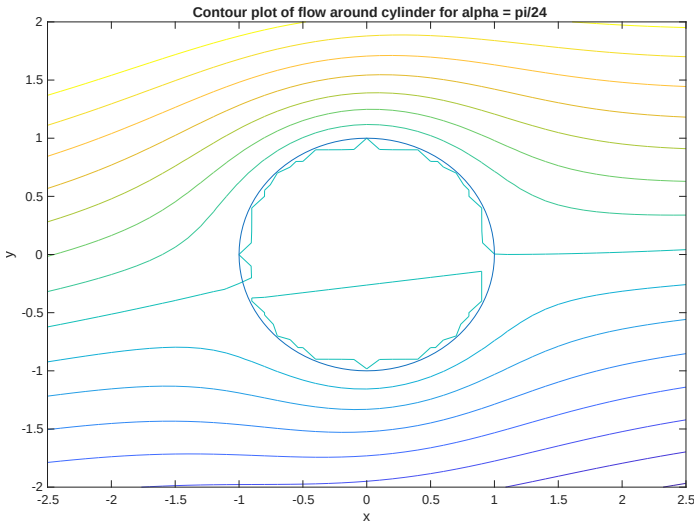
c = -1.75:0.25:1.75;

contour(xm,ym,psi,c);
xlabel('x');
ylabel('y');
title('Contour plot of flow around cylinder');

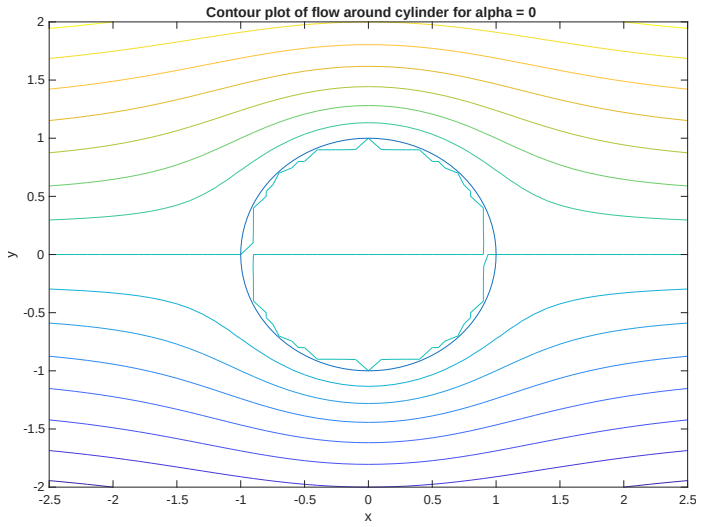
hold on
plot(xs,ys)
hold off
print -deps2c cylinder_flow_ex_4.eps;
```

5 Panel method solution for the cylinder flow

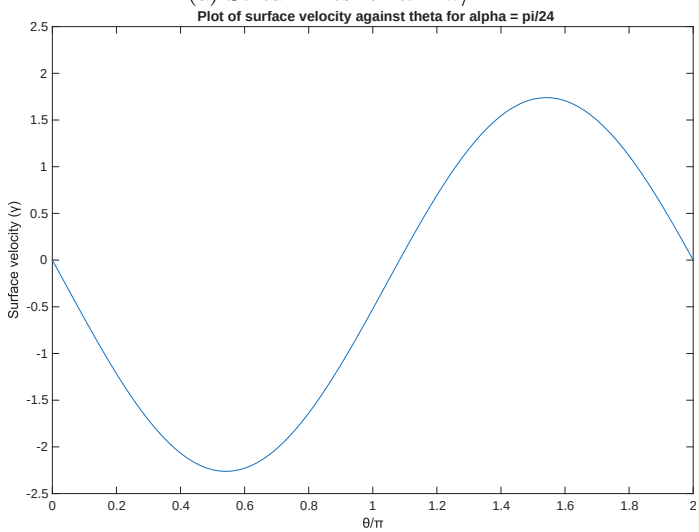
When $\alpha = \frac{\pi}{24}$ then the circulation is $\Gamma = -1.6411$.



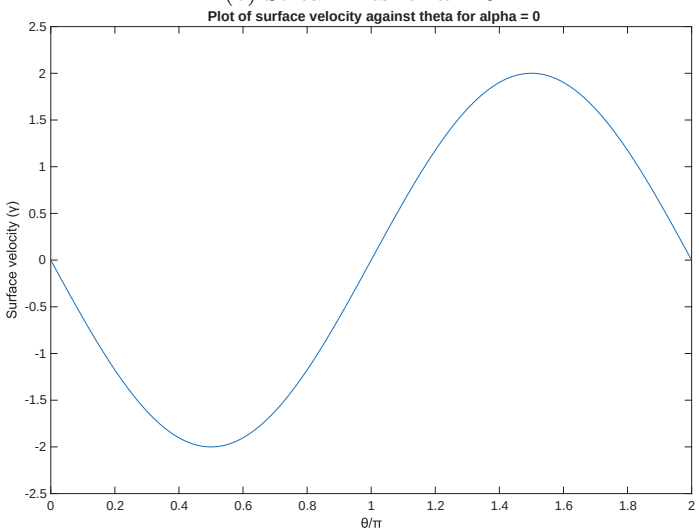
(a) Streamlines for $\alpha = \pi/24$



(b) Streamlines for $\alpha = 0$



(c) Surface velocity against angle for $\alpha = \pi/24$



(d) Surface velocity against angle for $\alpha = 0$

Figure 5: Cylinder flow solved using panel methods

Listing 8: Function for generating the **A** matrix to solve the panel method problem. (Exercise 5)

```
function lhsmat = build_lhs(xs, ys)
    np = length(xs)-1;
    psip = zeros(np, np+1);
    infa = zeros(np, np+1);
    infb = zeros(np, np+1);

    for i=1:np %special case for j=1
        [infa(i,1), infb(i,1)] = panelinf(xs(1), ys(1), xs(2), ys(2), xs(i), ys(i));
        psip(i, 1) = infa(i,1);
    end
    for j=2:np % main loop to generate psip matrix
        for i=1:np
            [infa(i, j), infb(i,j)] = panelinf(xs(j), ys(j), xs(j+1), ys(j+1), xs(i), ys(i));
            psip(i,j) = infa(i,j) + infb(i,j-1);
        end
    end
```

```
    for i =1:np %special case for j = np+1
        psip(i,np+1) = infb(i,j-1);
    end
    lhsmat = zeros(np+1, np+1);
    for i=1:np-1
        for j=1:np+1
            lhsmat(i,j) = psip(i+1, j) - psip(i,j);
        end
    end
    %set up the equation that gamma_np+1 = gamma_1
    lhsmat(np, 1) = 1;
    lhsmat(np, np+1) = -1;

    %set up the equation that gamma_1 = 0
    lhsmat(np+1,1)=1;

end
```

Listing 9: Function for generating the **b** vector for the panel method problem. (Exercise 5)

```
function rhsvec = build_rhs(xs,ys, alpha)
    np = length(xs)-1;
    rhsvec = zeros(np+1, 1); %column vector b
    psi_fs = zeros(np,1);

    for i = 1:np
        psi_fs(i,1) = ys(i)*cos(alpha) - xs(i)*sin(alpha); %free stream streamfunction
    end
    for i = 1:np-1
        rhsvec(i,1) = psi_fs(i,1)-psi_fs(i+1,1);
    end
    rhsvec(np,1)=0;
    rhsvec(np+1,1)=0;
    %keep b(np) and b(np+1)=0 such that gamma_0=1 and gamma_0 = gamma_np+1
end
```

Listing 10: Script for solving the panel method problem for a cylinder at an angle of attack. (Exercise 5)

```
clear;
xmin = -2.5;
xmax = 2.5;
ymin = -2;
ymax = 2;
ny = 41;
nx = 51;

psi = zeros(nx, ny);
psi_fs = zeros(nx,ny);
xm = zeros(nx, ny);
ym = zeros(nx, ny);
alpha = pi/24;

np = 100;
theta = (0:np)*2*pi/np;
xs = zeros(np+1);
ys = zeros(np+1);

for i = 1:np+1
    %points around the cylinder
    xs(i) = cos(theta(i));
    ys(i) = sin(theta(i));
end

```

```
%solve the panel method problem by generating matricies and using gam = A^-1 b
A = build_lhs(xs, ys);
b = build_rhs(xs, ys, alpha);
gam = A\b;


for i = 1:nx
    for j = 1:ny
        xm(i,j) = xmin + (i-1)*(xmax-xmin)/(nx-1);
        ym(i,j) = ymin + (j-1)*(ymax-ymin)/(ny-1);
        psi(i,j) = ym(i,j)*cos(alpha) - xm(i,j)*sin(alpha); %free stream steamfunction
        for s = 1:np
            %evaluate the influence coefficients and streamfunction
            [infa, infb] = panelinf(xs(s), ys(s), xs(s+1), ys(s+1), xm(i,j), ym(i,j));
            psi(i,j) = psi(i,j) + gam(s)*infa + gam(s+1) *infb;
        end
    end
end


c = -1.75:0.25:1.75;
circ =0;
for i=1:np
    %calculate the total circulation around the cylinder
    circ = circ + gam(i)*2*pi/np;
end

contour(xm,ym,psi,c);
xlabel('x');
ylabel('y');
title('Contour plot of flow around cylinder for alpha = pi/24');

hold on
plot(xs,ys)
hold off
print -deps2c contour_alpha_pi24.eps;


theta_pi = theta/pi;
figure(2);
plot(theta_pi,gam);
xlabel('theta/pi');
ylabel('Surface velocity (gamma)');
title('Plot of surface velocity against theta for alpha = pi/24')
axis([0 2 -2.5 2.5]);
print -deps2c sur_vel_alpha_pi24.eps;
```